

Mark Scheme — Practice Paper 3 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $(2x - 1)(x - 3) \leq 0$ (M1 factorise, A1 critical values $\frac{1}{2}, 3$). $\frac{1}{2} \leq x \leq 3$ (A1). [3]

2(a) $2(x - 2)^2 - 3$ (B1 $a = 2, b = -2$; B1 $c = -3$). [2]

2(b) Vertex $(2, -3)$ (B1 FT); line of symmetry $x = 2$ (B1 FT). [2]

3(a) Segment = $\frac{1}{2}r^2(\theta - \sin \theta) = \frac{1}{2} \times 36 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$ (M1) = $12\pi - 9\sqrt{3} \text{ cm}^2$ (A1). (≈ 22.1) [2]

3(b) Arc = $6 \times \frac{2\pi}{3} = 4\pi$; chord = $2 \times 6 \sin \frac{\pi}{3} = 6\sqrt{3}$ (M1 both). Perimeter = $4\pi + 6\sqrt{3} \text{ cm}$ (A1). (≈ 23.0) [2]

4(a) $1 + 6(-2x) + 15(-2x)^2$ (M1 binomial coefficients 6, 15) = $1 - 12x + 60x^2$ (A1 $-12x$, A1 $60x^2$). [3]

4(b) x^2 terms: $1 \times 60x^2 + 3x \times (-12x)$ (M1) $\Rightarrow 60 - 36 = 24$ (A1). [2]

5(a) Gradient of $l_1 = -\frac{2}{3}$ (B1), so gradient of $l_2 = \frac{3}{2}$ (M1). $y - 5 = \frac{3}{2}(x - 5) \Rightarrow 3x - 2y = 5$ (A1). [3]

5(b) Solve $2x + 3y = 12, 3x - 2y = 5$: $B = (3, 2)$ (M1 A1). $AB = \sqrt{(5-3)^2 + (5-2)^2} = \sqrt{13}$ (A1). [3]

6(a) $(x - 3)^2 + 2$ (B1 B1). [2]

6(b) $k = 3$ (B1). [1]

6(c) $y = (x - 3)^2 + 2 \Rightarrow x = 3 + \sqrt{y - 2}$ (M1, positive root since $x \geq 3$). $f^{-1}(x) = 3 + \sqrt{x - 2}$ (A1). Domain $x \geq 2$ (B1). [3]

7(a) LHS = $\frac{\cos \theta(1 + \sin \theta) - \cos \theta(1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}$ (M1) = $\frac{2 \sin \theta \cos \theta}{1 - \sin^2 \theta} = \frac{2 \sin \theta \cos \theta}{\cos^2 \theta}$ (M1) = $\frac{2 \sin \theta}{\cos \theta} = 2 \tan \theta$ (A1 AG). [3]

7(b) $2 \tan \theta = 3 \sin \theta \Rightarrow \frac{2 \sin \theta}{\cos \theta} - 3 \sin \theta = 0 \Rightarrow \sin \theta \left(\frac{2}{\cos \theta} - 3 \right) = 0$ (M1, must not divide by $\sin \theta$). $\sin \theta = 0 \Rightarrow \theta = 0^\circ, 180^\circ, 360^\circ$ (A1). $\cos \theta = \frac{2}{3} \Rightarrow \theta = 48.2^\circ$ (A1), 311.8° (A1). Five solutions in total. [4]

8(a) $\frac{dy}{dx} = 3 \times \frac{1}{2}(4x + 1)^{-1/2} \times 4 - 2$ (M1 chain rule, A1) = $\frac{6}{\sqrt{4x + 1}} - 2$ (A1). [3]

8(b) $\frac{6}{\sqrt{4x + 1}} = 2 \Rightarrow \sqrt{4x + 1} = 3 \Rightarrow x = 2$ (M1). $y = 3 \times 3 - 4 = 5$: point $(2, 5)$ (A1). [2]

8(c) $\frac{d^2y}{dx^2} = -12(4x + 1)^{-3/2}$ (M1), which is negative at $x = 2$ ($= -\frac{12}{27}$), so maximum (A1). [Gradient sign change either side is also acceptable.] [2]

9(a) $y = \int 12(2x + 1)^{-3} dx = \frac{12(2x + 1)^{-2}}{-2 \times 2} + c$ (M1 A1) = $-\frac{3}{(2x + 1)^2} + c$. At $(0, 1)$: $-3 + c = 1 \Rightarrow c = 4$ (M1). $y = 4 - \frac{3}{(2x + 1)^2}$ (A1). [4]

9(b) Curve crosses x -axis at $x = 0$ and $x = 2$; the region for $0 \leq x \leq 2$ is below the axis (B1). $\int_0^2 (x^2 -$

$2x) dx = [\frac{1}{3}x^3 - x^2]_0^2 = -\frac{4}{3}$, so area $\frac{4}{3}$ (M1 A1). $\int_2^3 (x^2 - 2x) dx = (9 - 9) - (\frac{8}{3} - 4) = \frac{4}{3}$. Total area $= \frac{8}{3}$ (A1). [A single integral $\int_0^3$ giving 0 scores at most B1.] [4]

10(a) $12 + 7d = 33 \Rightarrow d = 3$ (B1). $\frac{n}{2}(24 + 3(n - 1)) = 495$ (M1) $\Rightarrow 3n^2 + 21n - 990 = 0 \Rightarrow n^2 + 7n - 330 = 0$ (A1) $\Rightarrow (n - 15)(n + 22) = 0 \Rightarrow n = 15$ (A1). [4]

10(b)(i) $S_\infty = \frac{50}{1 - 0.8} = 250$ (B1). [1]

10(b)(ii) $S_{10} - S_4$ (M1) $= \frac{50(1 - 0.8^{10})}{0.2} - \frac{50(1 - 0.8^4)}{0.2}$ (A1) $= 223.16 - 147.6 = 75.6$ (A1).

[Alternatively $u_5 \frac{1 - 0.8^6}{1 - 0.8}$ with $u_5 = 20.48$.] [3]

11(a) Centre = midpoint $(2, 6)$ (B1). Radius $= \frac{1}{2}AB = \frac{1}{2}\sqrt{36 + 64} = 5$ (M1). $(x - 2)^2 + (y - 6)^2 = 25$ (A1). [3]

11(b) $(6 - 2)^2 + (3 - 6)^2 = 16 + 9 = 25$ ✓ (B1). Gradient of radius $= \frac{3-6}{6-2} = -\frac{3}{4}$, tangent gradient $\frac{4}{3}$ (M1). $y - 3 = \frac{4}{3}(x - 6) \Rightarrow 4x - 3y = 15$ (A1). [3]

11(c) $y = 0 \Rightarrow x = \frac{15}{4}$: $D(\frac{15}{4}, 0)$ (B1). $CD = \sqrt{(6 - \frac{15}{4})^2 + 3^2} = \sqrt{\frac{81}{16} + 9} = \sqrt{\frac{225}{16}}$ (M1) $= \frac{15}{4} = 3.75$ (A1). [3]

12(a) $-3 \leq f(x) \leq 1$ (B1 for -3 , B1 for 1 ; must be an inequality in $f(x)$ or y). [2]

12(b) Cosine shape, one full cycle from $x = 0$ to 2π , starting at $(0, 1)$ and ending at $(2\pi, 1)$ (B1); minimum -3 at $x = \pi$, maxima at $x = 0, 2\pi$ clearly marked (B1). [2]

12(c) Stretch parallel to the y -axis, scale factor 2 (B1); then translation by $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$ (B1). (Order matters: translation first would give $2 \cos x - 2$.) [2]

12(d) $\cos x = \frac{1}{2}$ (M1). $x = \frac{\pi}{3}, \frac{5\pi}{3}$ (A1). [2]

Total: 75